



AI reinvention made real

Accenture Innovation Challenge 2026

Team details

TEAM NAME: Team cognition



Saurav Gupta(Team Leader)

College: IIT Jodhpur
Stream: CSE
Year of graduation: 2027



Abhishek Gehlot

College: IIT Jodhpur
Stream: CSE
Year of graduation: 2027

Describe the problem statement (200 words)

Every enterprise today drowns in dashboards. A BI tool can instantly show that APAC revenue dropped **8% last Tuesday** but it cannot explain *why*, or prescribe *what to do next*. That critical "last mile" of analytics transforming raw metric fluctuations into executive-grade root-cause narratives and concrete business actions still depends entirely on human analysts manually cross-referencing SQL queries, Jira deployments, Zendesk tickets, and Slack threads. This bottleneck delays actionable insights by **3–5 business days**, costing enterprises millions in slow response time.

The challenge demands building a KPI Storytelling Engine an autonomous AI system that ingests both structured time-series metrics and unstructured enterprise data to generate natural-language diagnostic narratives explaining metric movements, identifying causal root factors, and recommending prioritized next steps.

Three fundamental analytical hurdles make this non-trivial. First, the engine must statistically distinguish genuine **operational anomalies from normal seasonal variance and daily noise**. Second, it must graduate beyond surface-level data correlations to deliver causally-grounded, SOP-backed prescriptive actions that executives can immediately execute. Third and most critically it must possess epistemic self-awareness: recognizing when evidence is genuinely ambiguous and recommending targeted diagnostic experiments instead of hallucinating false conclusions.

Proposed solution (200 words)

Our solution, **CortexKPI**, is a four-layer autonomous storytelling engine that transforms raw metric anomalies into executive-ready causal narratives with zero human analyst intervention.

Layer 1 — Noise-Aware Anomaly Detection: Implements Bayesian dynamic baselining, computing rolling 28-day confidence intervals for every KPI. The system only triggers deep analysis when a metric mathematically breaches the 95% statistical boundary ($|Z\text{-score}| > 2.0$), eliminating false alarms from seasonal variance and daily jitter.

Layer 2 — Causal Metric Tree Decomposition: Traverses a directed metric dependency graph (Revenue $\rightarrow$ Sessions $\times$ Conversion $\times$ AOV) top-down, mathematically isolating the exact failing sub-metric branch rather than reporting shallow correlations.

Layer 3 — Multimodal Evidence Fusion: Employs **Retrieval-Augmented Generation** across embedded unstructured enterprise sources Jira deployments, Zendesk customer tickets, Slack engineering alerts semantically correlating operational events with the anomaly's exact time window to corroborate the structured diagnosis with real-world qualitative evidence.

Layer 4 — Prescriptive Synthesis with Epistemic Safeguards: Generates actionable, **SOP-backed executive narratives** with prioritized interventions. Crucially, an internal confidence scorer activates an **Honest Detective** protocol when evidence is weak or contradictory transparently flagging ambiguity and recommending targeted diagnostic experiments (micro-surveys, A/B tests) instead of hallucinating false conclusions.

https://drive.google.com/drive/folders/11dQg3_jTakLCV-3CgUoHqiAmDaAbrXIX?usp=sharing

Thank you